

Guselkumab (Tremfya)

Policy History

- Last Review 07/21/2025

Effective: 12/16/2022

Next Review: 03/26/2026

- Review History
- Definitions

Additional Information

- Clinical Policy Bulletin Notes

Number: 1011

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Scope of Policy

This Clinical Policy Bulletin addresses guselkumab (Tremfya) for commercial medical plans. For Medicare criteria, see Medicare Part B Criteria.

Note: Requires Precertification:

Precertification of intravenous guselkumab (Tremfya IV) is required of all Aetna participating providers and members in applicable plan designs. For precertification of intravenous guselkumab (Tremfya IV), call (866) 752-7021 or fax (888) 267-3277. For Statement of Medical Necessity (SMN) precertification forms, see Specialty Pharmacy Precertification.

1. Prescriber Specialties

This medication must be prescribed by or in consultation with *one* of the following:

1. Plaque psoriasis: dermatologist;
2. Psoriatic arthritis: rheumatologist or dermatologist;
3. Ulcerative colitis and Crohn's disease: gastroenterologist.

2. Criteria for Initial Approval

Aetna considers guselkumab (Tremfya) medically necessary for the following indications when criteria are met:

1. Plaque psoriasis (PsO)

1. For adult members who have previously received a biologic or targeted synthetic drug (e.g., Sotyktu, Otezla) indicated for the treatment of moderate to severe plaque psoriasis; *or*
2. For adult members for treatment of moderate-to-severe plaque psoriasis when *any* of the following criteria is met:

1. Crucial body areas (e.g., hands, feet, face, neck, scalp, genitals/groin, intertriginous areas) are affected;
or
2. At least 10% of the body surface area (BSA) is affected; *or*
3. At least 3% of the BSA is affected and the member meets *either* of the following criteria:
 1. Member has had an inadequate response or intolerance to either phototherapy (e.g., UVB, PUVA) or pharmacologic treatment with methotrexate, cyclosporine, or acitretin; *or*
 2. Member has a clinical reason to avoid pharmacologic treatment with methotrexate, cyclosporine, and acitretin (see Appendix);

2. Psoriatic arthritis (PsA)

1. For adult members who have previously received a biologic or targeted synthetic drug (e.g., Rinvoq, Otezla) indicated for active psoriatic arthritis; *or*
2. For adult members for treatment of active psoriatic arthritis when *either* of the following criteria is met:
 1. Member has mild to moderate disease and meets *one* of the following criteria:
 1. Member has had an inadequate response to methotrexate, leflunomide, or another conventional synthetic drug (e.g., sulfasalazine) administered at an adequate dose and duration; *or*
 2. Member has an intolerance or contraindication to methotrexate or leflunomide (see Appendix), or another conventional synthetic drug (e.g., sulfasalazine); *or*
 3. Member has enthesitis; *or*
 2. Member has severe disease;

3. Ulcerative colitis (UC)

For treatment of moderately to severely active ulcerative colitis;

4. Crohn's disease (CD)

For treatment of moderately to severely active Crohn's disease.

Aetna considers all other indications as experimental, investigational, or unproven.

3. Continuation of Therapy

Aetna considers continuation of guselkumab (Tremfya) therapy medically necessary for the following indications:

1. Plaque psoriasis (PsO)

For all adult members (including new members) who are using the requested medication for moderate to severe plaque psoriasis and who achieve or maintain positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when *either* of the following is met:

1. Reduction in body surface area (BSA) affected from baseline; *or*
2. Improvement in signs and symptoms from baseline (e.g., itching, redness, flaking, scaling, burning, cracking, pain);

2. Psoriatic arthritis (PsA)

For all adult members (including new members) who are using the requested medication for psoriatic arthritis and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when there is improvement in *any* of the following from baseline:

1. Number of swollen joints; *or*
2. Number of tender joints; *or*
3. Dactylitis; *or*
4. Enthesitis; *or*
5. Skin and/or nail involvement; *or*
6. Functional status; *or*

7. C-reactive protein (CRP);

3. Ulcerative colitis

1. For all members (including new members) who are using the requested medication for moderately to severely active ulcerative colitis and who achieve or maintain remission; or
2. For all members (including new members) who are using the requested medication for moderately to severely active ulcerative colitis and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when there is improvement in *any* of the following from baseline:
 1. Stool frequency; *or*
 2. Rectal bleeding; *or*
 3. Urgency of defecation; *or*
 4. C-reactive protein (CRP); *or*
 5. Fecal calprotectin (FC); *or*
 6. Appearance of the mucosa on endoscopy, computed tomography enterography (CTE), magnetic resonance enterography (MRE), or intestinal ultrasound; *or*
 7. Improvement on a disease activity scoring tool (e.g., Ulcerative Colitis Endoscopic Index of Severity [UCEIS], Mayo score);

4. Crohn's disease (CD)

1. For all members (including new members) who are using the requested medication for moderately to severely active Crohn's disease and who achieve or maintain remission; *or*
2. For all members (including new members) who are using the requested medication for moderately to severely active Crohn's disease and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when there is improvement in *any* of the following from baseline:
 1. Abdominal pain or tenderness; *or*
 2. Diarrhea; *or*
 3. Body weight; *or*
 4. Abdominal mass; *or*
 5. Hematocrit; *or*
 6. Appearance of the mucosa on endoscopy, computed tomography enterography (CTE), magnetic resonance enterography (MRE), or intestinal ultrasound; *or*
 7. Improvement on a disease activity scoring tool (e.g., Crohn's Disease Activity Index [CDAI] score).

4. Other

For all indications: Member has had a documented negative tuberculosis (TB) test (which can include a tuberculosis skin test [TST] or an interferon-release assay [IGRA])^{Footnote1*} within 12 months of initiating therapy for persons who are naïve to biologic drugs or targeted synthetic drugs associated with an increased risk of TB.

^{Footnote1*} If the screening testing for TB is positive, there must be further testing to confirm there is no active disease (e.g., chest x-ray). Do not administer the requested medication to members with active TB infection. If there is latent disease, TB treatment must be started before initiation of the requested medication.

For all indications: Member cannot use the requested medication concomitantly with any other biologic drug or targeted synthetic drug for the same indication.

5. Related Policies

1. CPB 0205 - Phototherapy and Photochemotherapy (PUVA) for Skin Conditions
2. CPB 0315 - Etanercept
3. CPB 0341 - Infliximab
4. CPB 0577 - Laser Treatment for Psoriasis and Other Selected Skin Conditions
5. CPB 0655 - Adalimumab
6. CPB 0720 - Abatacept (Orencia)
7. CPB 0761 - Certolizumab Pegol (Cimzia)
8. CPB 0790 - Golimumab (Simponi and Simponi Aria)
9. CPB 0885 - Vedolizumab (Entyvio)
10. CPB 0905 - Secukinumab (Cosentyx)

11. CPB 0912 - Ustekinumab
12. CPB 1009 - Risankizumab-rzaa (Skyrizi)

Dosage and Administration

Note: Approvals may be subject to dosing limits in accordance with FDA-approved labeling, accepted compendia, and/or evidence-based practice guidelines.

Guselkumab is available as Tremfya for subcutaneous injection or intravenous infusion.

Tremfya is supplied in the following dosage forms and strengths:

- Subcutaneous injection:
 - Injection: 100 mg/mL in a single-dose One-Press patient-controlled injector
 - Injection: 100 mg/mL in a single-dose prefilled pen (Tremfya Pen)
 - Injection: 200 mg/2 mL in a single-dose prefilled pen (Tremfya Pen)
 - Injection: 100 mg/mL in a single-dose prefilled syringe
 - Injection: 200 mg/2 mL (100 mg/mL) in a single-dose prefilled syringe
- Intravenous infusion:
 - Injection: 200 mg/20 mL (10 mg/mL) solution in a single-dose vial.

Subcutaneous injection of Tremfya may be administered by a health care professional, or a person may self-inject after proper training in subcutaneous injection technique. Tremfya solution for intravenous infusion must be diluted, prepared, and infused by a healthcare professional using aseptic technique.

Plaque Psoriasis (PsO)

The recommended dose is 100 mg administered as a subcutaneous injection at Week 0, Week 4, and every 8 weeks thereafter.

Psoriatic Arthritis (PsA)

The recommended dose is 100 mg administered as a subcutaneous injection at Week 0, Week 4, and every 8 weeks thereafter. May be administered alone or in combination with a conventional disease-modifying antirheumatic drug (cDMARD) (e.g., methotrexate).

Ulcerative Colitis (UC)

Induction: 200 mg is administered by intravenous infusion over at least one hour at Week 0, Week 4, and Week 8.

Maintenance: 100 mg is administered by subcutaneous injection at Week 16, and every 8 weeks thereafter, or 200 mg is administered by subcutaneous injection at Week 12, and every 4 weeks thereafter. Use the lowest effective recommended dosage to maintain therapeutic response.

Crohn's Disease (CD)

Induction: 200 mg is administered by intravenous infusion over at least one hour at Week 0, Week 4, and Week 8; or 400 mg administered by subcutaneous injection at Week 0, Week 4, and Week 8.

Maintenance: 100 mg is administered by subcutaneous injection at Week 16, and every 8 weeks thereafter; or 200 mg administered by subcutaneous injection at Week 12, and every 4 weeks thereafter. The lowest effective recommended dosage is used to maintain therapeutic response.

Source: Janssen Biotech, 2025

Experimental, Investigational, or Unproven

Aetna considers concomitant use of guselkumab with any other biologic drug (e.g., adalimumab, anakinra, etanercept, infliximab, rilonacept, tocilizumab) or targeted synthetic drug (e.g. tofacitinib) experimental, investigational, or unproven for the same indication because the effectiveness of this approach has not been established.

Table:

CPT Codes / HCPCS Codes / ICD-10 Codes

Other CPT codes related to the CPB:

Information in the [brackets] below has been added for clarification purposes. Codes requiring a 7th character are represented by "+":

Code	Code Description
71045 -hyphen 71048	Radiologic examination, chest
86140	C-hyphenreactive protein
86141	C-hyphenreactive protein; high sensitivity (hsCRP)
86480	Tuberculosis test, cell mediated immunity antigen response measurement; gamma interferon
86481	enumeration of gamma interferon -hyphen producing T cells in cell suspension
86580	Skin test; tuberculosis, intradermal
96372	Therapeutic, prophylactic, or diagnostic injection (specify substance or drug); subcutaneous or intramuscular
96910	Photochemotherapy; tar and ultraviolet B (Goeckerman treatment) or petrolatum and ultraviolet B
96912	psoralens and ultraviolet A (PUVA)
96913	Photochemotherapy (Goeckerman and/or PUVA) for severe photoresponsive dermatoses requiring at least 4-hyphen8 hours of care under direct supervision of the physician (includes application of medication and dressings)

HCPCS codes covered if selection criteria are met:

J1628	Injection, guselkumab, 1 mg
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Other HCPCS codes related to the CPB:

Sotyktu, Rinvoq, Otezla -hyphenno specific code

J0139	Injection, adalimumab, 1 mg
J1438	Injection, etanercept, 25 mg
J1745	Injection infliximab, 10 mg
J3262	Injection, tocilizumab, 1 mg
J7502	Cyclosporine, oral, 100 mg
J7515	Cyclosporine, oral, 25 mg
J7516	Cyclosporine, parenteral, 250 mg
J8610	Methotrexate, oral, 2.5 mg
J8611	Methotrexate (jylamvo), oral, 2.5 mg
J8612	Methotrexate (xatmep), oral, 2.5 mg
J9250	Methotrexate sodium, 5 mg
J9255	Injection, methotrexate (accord) not therapeutically equivalent to j9250 or j9260, 50 mg
J9260	Methotrexate sodium, 50 mg
Q5103	Injection, infliximab-hyphenhyphenndyyb, biosimilar, (inflectra), 10 mg
Q5104	Injection, infliximab-hyphenhyphenabda, biosimilar, (renflexis), 10 mg
Q5109	Injection, infliximab-hyphenqbtx, biosimilar, (ixifi), 10 mg

Information in the [brackets] below has been added for clarification purposes. Codes requiring a 7th character are represented by "+":

Code	Code Description
Q5121	Injection, infliximab-hyphenhyphenaxxq, biosimilar, (avsola), 10 mg
Q5133	Injection, tocilizumab-hyphenbavi (tofidence), biosimilar, 1 mg
Q5135	Injection, tocilizumab-hyphenaaazg (tyenne), biosimilar, 1 mg
Q5140	Injection, adalimumab-hyphenfkjp, biosimilar, 1 mg
Q5141	Injection, adalimumab-hyphenaaty, biosimilar, 1 mg
Q5142	Injection, adalimumab-hyphenryvk biosimilar, 1 mg
Q5143	Injection, adalimumab-hyphenadbm, biosimilar, 1 mg
Q5144	Injection, adalimumab-hyphenaaaf (idacio), biosimilar, 1 mg
Q5145	Injection, adalimumab-hyphenafzb (abrilada), biosimilar, 1 mg

ICD-hyphen10 codes covered if selection criteria are met:

K50.00 – K50.919	Crohn's disease [regional enteritis]
K51.00 - hyphen K51.919	Ulcerative colitis
L40.0 -hyphen L40.9	Psoriasis

Background

U.S. Food and Drug Administration (FDA)-Approved Indications

- Treatment of adult patients with moderate-to-severe plaque psoriasis (PsO) who are candidates for systemic therapy or phototherapy
- Treatment of adult patients with active psoriatic arthritis (PsA)
- Treatment of moderately to severely active ulcerative colitis (UC) in adults
- Treatment of moderately to severely active Crohn's disease (CD) in adults

Guselkumab, branded as Tremfya (Janssen Biotech, Inc), is a human monoclonal antibody that selectively binds to the p19 subunit of interleukin 23 (IL-23), effectively blocking its interaction with the IL-23 receptor. This prevents the release of proinflammatory cytokines and chemokines, as IL-23 is a naturally occurring cytokine that plays a key role in inflammatory and immune responses.

Labeled warning and precautions for guselkumab include hypersensitivity reactions, infections, and hepatotoxicity. Serious hypersensitivity reactions, including anaphylaxis, have been reported with post-market use of Tremfya. Some cases required hospitalization. In addition, Tremfya may increase the risk of infection. In clinical trials in subjects with plaque psoriasis, infections occurred in 23% of subjects in the Tremfya group versus 21% of subjects in the placebo group through 16 weeks of treatment. Upper respiratory tract infections, gastroenteritis, tinea infections, and herpes simplex infections occurred more frequently in the Tremfya group than in the placebo group. A similar risk of infection was seen in placebo-controlled trials in subjects with psoriatic arthritis. Moreover, a serious adverse reaction of drug-induced liver injury was reported in a clinical trial participant with Crohn's disease following three doses of a higher than the recommended induction regimen.

Prior to initiating treatment with Tremfya, patients should be evaluated for tuberculosis (TB) infection. Patients are not to receive Tremfya if they have active TB infection. Treatment for latent TB should be initiated prior to administering Tremfya. Per the prescribing information, consider anti-TB therapy prior to initiating Tremfya in patients with a past history of latent or active TB in whom an adequate course of treatment cannot be confirmed. Patients are to be monitored for signs and symptoms of active TB during and after Tremfya treatment. In clinical trials, 105 patients with plaque psoriasis, 71 patients with psoriatic arthritis, 31 patients with ulcerative colitis, and 36 patients with Crohn's disease with latent TB who were concurrently treated with guselkumab, and appropriate TB prophylaxis did not develop active TB. In clinical trials of guselkumab in patients with Crohn's disease, active TB was reported in 2 patients during treatment with guselkumab.

Per the Centers for Disease Control and Prevention (CDC), the tuberculin (TB) skin test (also called the Mantoux tuberculin skin test [TST]) and the TB blood test (also called interferon-gamma release assays or IGRAs) can be used to test for M. tuberculosis

infection. Additional tests are required to confirm TB disease. "A posterior-anterior chest radiograph is used to detect chest abnormalities. Lesions may appear anywhere in the lungs and may differ in size, shape, density, and cavitation. These abnormalities may suggest TB, but cannot be used to definitively diagnose TB. However, a chest radiograph may be used to rule out the possibility of pulmonary TB in a person who has had a positive reaction to a TST or TB blood test and no symptoms of disease" (CDC, 2024).

Live vaccines should be avoided in patients treated with guselkumab, as medications that interact with the immune system may increase the risk of infection following administration of live vaccines.

The most common adverse reactions associated with Tremfya include:

- Plaque Psoriasis and Psoriatic Arthritis (1% or more): upper respiratory infections, headache, injection site reactions, arthralgia, bronchitis, diarrhea, gastroenteritis, tinea infections, and herpes simplex infections;
- Ulcerative Colitis
 - Induction (2% or more): respiratory tract infections;
 - Maintenance (3% or more): injection site reactions, arthralgia, and upper respiratory tract infection;
- Crohn's Disease (3% or more): respiratory tract infections, abdominal pain, injection site reactions, headache, fatigue, arthralgia, diarrhea, and gastroenteritis.

Crohn's Disease

In March 2025, the U.S. Food and Drug Administration (FDA) approved Tremfya (guselkumab) for the treatment of adults with moderately to severely active Crohn's disease (CD), a chronic inflammatory condition of the gastrointestinal tract. This approval includes both subcutaneous (SC) and intravenous (IV) induction options. FDA approval is based on results from multiple Phase 3 trials evaluating more than 1,300 patients with moderately to severely active CD who failed or were intolerant to conventional therapy (i.e. corticosteroids or immunomodulators) or biologics. Moderately to severely active CD was defined as a Crohn's Disease Activity Index (CDAI) score of greater than or equal to 220 and a Simple Endoscopic Score for Crohn's Disease (SES-CD) of greater than or equal to 6 (or greater than or equal to 4 for patients with isolated ileal disease). Patients were permitted to use stable doses of oral corticosteroids (prednisone less than or equal to 40 mg/day or equivalent), immunomodulators (azathioprine, 6-mercaptopurine, methotrexate), and/or aminosalicylates. The comprehensive results from these studies demonstrated significant clinical remission and endoscopic response at 48 weeks with both SC or IV induction routes.

The GALAXI trial (ClinicalTrials.gov ID: NCT03466411) is a randomized, double-blind, placebo-controlled, active-controlled (ustekinumab), global, multicenter Phase 2/3 program designed to evaluate the efficacy and safety of guselkumab in participants with moderately to severely active CD with inadequate response/intolerance to conventional therapies (corticosteroids or immunomodulators) and/or biologics (TNF antagonists or vedolizumab). The GALAXI trial included a Phase 2 dose-ranging study (GALAXI 1) and two independent, identically designed confirmatory Phase 3 studies (GALAXI 2 and 3, n=1021). Patients received guselkumab 200 mg IV induction at Weeks 0, 4 and 8 followed by guselkumab 200 mg SC maintenance every 4 weeks; or guselkumab 200 mg IV induction at Weeks 0, 4 and 8, followed by guselkumab 100 mg SC maintenance every 8 weeks; or a biologic active control; or placebo. Participants randomized to placebo were able to receive ustekinumab if clinical response was not met at Week 12. Of the 873 individuals pooled across the GALAXI 2 & 3 dataset, 456 (52 percent) had prior history of inadequate response to biologics, 365 (42 percent) were biologic-naïve and 52 (6 percent) were biologic experienced without documented inadequate response or intolerance. Each GALAXI study employed a treat-through design in which participants remained on the treatment to which they were initially randomized and includes a long-term extension study that will assess clinical, endoscopic, and safety outcomes with guselkumab through a total of 5 years.

The results of the GALAXI 2 and 3 trials found that at Week 12 for IV induction with guselkumab 200 mg at Weeks 0, 4, and 8, patients achieved clinical remission (47% in both trial) verses placebo (20% and 15%) $p<0.001$; and achieved endoscopic response (36% and 34%) verses placebo (9% and 13%) ($p<0.001$).

The GRAVITI trial (ClinicalTrials.gov ID: NCT05197049) is a randomized, double-blind, placebo-controlled Phase 3 study that evaluated guselkumab SC induction therapy (400 mg at Weeks 0, 4, and 8) in 347 patients with moderately to severely active CD who experienced an inadequate response or failed to tolerate conventional therapy (i.e., corticosteroids or immunomodulators) or biologic therapy (TNF antagonists or vedolizumab). Patients received guselkumab 400 mg SC every 4 weeks (x3) followed by guselkumab 200 mg SC every 4 weeks; or guselkumab 400 mg SC every 4 weeks (q4w) (x3) followed by guselkumab 100 mg SC every 8 weeks (q8w); or placebo. The maintenance doses in the GRAVITI trial (200 mg SC q4w and 100 mg SC q8w) were the same as those evaluated in the Phase 3 GALAXI 2 and GALAXI 3 studies that evaluated the efficacy and safety of IV induction followed by SC maintenance therapy in patients with moderate to severely active CD. Similar to GALAXI, GRAVITI employed a treat-through design, in which patients were randomized to guselkumab at Week 0 and remained on that regimen (from initial randomization) throughout the study, regardless of clinical response status at the end of induction. The coprimary endpoints were clinical remission and endoscopic response at Week 12 compared to placebo. Endoscopic remission was defined as an Simple Endoscopic Score - Crohn's disease (SES-CD) score less than or equal to 4 and at least a

2-point reduction from baseline and no subscore greater than 1 in any individual component. At Week 12 for SC induction with guselkumab 400 mg at Weeks 0, 4, and 8, clinical remission was achieved (56 percent) versus placebo (22 percent; $p < 0.001$); and endoscopic response was achieved (34 percent) versus placebo (15 percent; $p < 0.001$). At Week 48, a greater proportion of patients treated with either guselkumab regimen (i.e., SC guselkumab 400 mg at Weeks 0, 4, and 8 followed by SC guselkumab 100 mg at Week 16 and every 8 weeks thereafter or SC guselkumab 400 mg at Weeks 0, 4, and 8 followed by SC guselkumab 200 mg at Week 12 and every 4 weeks thereafter) achieved endoscopic remission, compared to placebo-treated patients (31% and 40%, respectively, vs. 6%). Participants randomized to placebo were able to receive guselkumab (400 mg SC q4w x3 to 100 mg SC q8w) if rescue criteria were met at Week 16.

In summary, the outcomes from the GRAVITI trial found that subcutaneous guselkumab induction induced clinical remission and endoscopic response in a significantly greater proportion of patients with moderately to severely active CD versus placebo. This trial includes a long-term extension that will assess clinical, endoscopic, and safety outcomes with guselkumab through a total of 5 years.

Plaque Psoriasis

In July 2017, the U.S. FDA approved Tremfya (guselkumab) for the treatment of moderate-to-severe plaque psoriasis in adults who are candidates for systemic therapy or phototherapy. FDA approval was based on results from Phase 3, multicenter, randomized, double-blind trials (VOYAGE 1, VOYAGE 2, and NAVIGATE studies). In VOYAGE 1 and VOYAGE 2, 1443 subjects, 18 years and older, were randomized to either Tremfya (100 mg at Weeks 0 and 4 and every 8 weeks thereafter), placebo or adalimumab (80 mg at Week 0 and 40 mg at Week 1, followed by 40 mg every other week thereafter). At 16 weeks, at least seven out of ten Tremfya-treated patients achieved at least 90 percent clearer skin, and more than 80 percent demonstrated cleared or almost cleared skin. Improvements were also demonstrated in psoriasis involving the scalp and in symptoms of plaque psoriasis including itch, pain, stinging, burning and skin tightness. Treatment resulted in clearer skin that lasted, as nearly nine out of ten patients who achieved PASI 90 at week 28 maintained that response at week 48. At week 24, more than seven out of ten patients treated with Tremfya reported at least 90 percent clearer skin compared with more than four out of ten patients treated with adalimumab.

NAVIGATE findings demonstrated the effectiveness of Tremfya in patients who had an inadequate response to treatment with ustekinumab. NAVIGATE evaluated the efficacy of 24 weeks of treatment with Tremfya in subjects (N=268) who had not achieved an adequate response, defined as IGA ≥ 2 at Week 16 after initial treatment with ustekinumab (dosed 45 mg or 90 mg according to the subject's baseline weight at Week 0 and Week 4). These subjects were randomized to either continue with ustekinumab treatment every 12 weeks or switch to Tremfya 100 mg at Weeks 16, 20, and every 8 weeks thereafter. In subjects with an inadequate response (IGA ≥ 2 at Week 16 to ustekinumab), greater proportions of subjects on Tremfya compared to ustekinumab achieved an IGA score of 0 or 1 with a ≥ 2 grade improvement at Week 28 (31% vs 14%, respectively; 12 weeks after randomization).

Psoriatic Arthritis

In July 2020, the U.S. FDA approved Tremfya (guselkumab) for the treatment of adult patients with active psoriatic arthritis (PsA), a chronic progressive disease characterized by painful joints and skin inflammation. FDA approval was based on the results from two pivotal Phase 3 clinical trials (DISCOVER-1 and DISCOVER-2), which were randomized, double-blind, placebo-controlled studies that evaluated the safety and efficacy of guselkumab in 1,120 adult patients with active PsA who had inadequate response to standard therapies.

Deodhar et al (2020) state that guselkumab, a specific inhibitor of interleukin-23 (IL-23) via IL-23 p19 subunit binding, significantly improved psoriatic arthritis signs and symptoms with an acceptable safety profile in a phase 2 trial. Thus, the authors conducted a multicenter, double-blind, randomized, placebo-controlled, phase 3 trial (DISCOVER-1) to evaluate guselkumab in adults with active psoriatic arthritis. Eligibility criteria included inadequate response to or intolerance of standard treatment, including at least 4 months of apremilast, at least 3 months of non-biologic disease-modifying antirheumatic drugs (DMARDs), or at least 4 weeks of non-steroidal anti-inflammatory drugs for psoriatic arthritis. About 30% of study participants could have previously received one or two TNF inhibitors. There were 381 patients randomly assigned (1:1:1) to subcutaneous guselkumab 100 mg every 4 weeks (n=128), guselkumab 100 mg at weeks 0, 4 and then every 8 weeks (n=127), or placebo (n=126). 362 patients continued study treatment up to week 24. The primary endpoint was American College of Rheumatology 20% improvement (ACR20) at week 24 in all patients per assigned treatment group using non-responder imputation. Safety was assessed in all patients per treatment received. The primary endpoint was met: ACR20 at week 24 was achieved by significantly greater proportions of patients in the guselkumab every 4 weeks group and every 8 weeks group than in the placebo group, with percentage differences versus placebo of 37% for the every 4 weeks group and 30% for the every 8 weeks group (both $p < 0.0001$). Serious adverse events up to week 24 occurred in no patients receiving guselkumab every 4 weeks, four (3%) patients receiving guselkumab every 8 weeks, and five (4%) patients receiving placebo. Up to week 24, one patient in the placebo group died from cardiac failure and two had serious infections; no guselkumab-treated patient died or had serious infections. The authors concluded that guselkumab demonstrated a favorable benefit-risk profile and might be an effective treatment option for patients with active psoriatic arthritis.

Mease et al (2020) conducted a multicenter, phase 3, double-blind, placebo-controlled study (DISCOVER-2) to evaluate guselkumab in biologic-naïve patients with active psoriatic arthritis (at least five swollen joints, at least five tender joints, and C-reactive protein ≥ 0.6 mg/dL) despite standard therapies. Patients were randomly assigned (1:1:1) to subcutaneous injections of guselkumab 100 mg every 4 weeks; guselkumab 100 mg at weeks 0, 4, then every 8 weeks; or placebo. The primary endpoint was ACR20 response at week 24 in all patients per assigned treatment group. Safety was assessed in all patients per treatment received. A total of 1153 patients were screened, of whom 741 were randomly assigned to receive guselkumab every 4 weeks (n=246), every 8 weeks (n=248), or placebo (n=247). One patient in the every 4 weeks group and one in the placebo group did not start treatment, and the remaining 739 patients started treatment; 716 patients continued treatment up to week 24. Significantly greater proportions of patients in the guselkumab every 4 weeks group (156 [64%] of 245 [95% CI 57-70]) and every 8 weeks group (159 [64%] of 248 [58-70]) than in the placebo group (81 [33%] of 246 [27-39]) achieved an ACR20 response at week 24 (percentage differences vs placebo 31% [95% CI 22-39] for the every 4 weeks group and 31% [23-40] for the every 8 weeks group; both $p < 0.0001$). Up to week 24, serious adverse events occurred in eight (3%) of 245 patients receiving guselkumab every 4 weeks (three serious infections), three (1%) of 248 receiving guselkumab every 8 weeks (one serious infection), and seven (3%) of 246 receiving placebo (one serious infection). No deaths occurred. The authors concluded that guselkumab was efficacious and demonstrated an acceptable benefit-risk profile in patients with active psoriatic arthritis who were naïve to treatment with biologics.

Ulcerative Colitis

In September 2024, Tremfya was granted FDA approval for the treatment of adults with moderately to severely active ulcerative colitis (UC), a chronic disease of the large intestine in which the lining of the colon becomes inflamed. Approval was based on the safety and efficacy results from the QUASAR study which met its primary endpoint at week 12, 22.6% of patients taking guselkumab achieved clinical remission, compared to 7.9% of patients taking a placebo.

QUASAR is a randomized, double-blind, placebo-controlled, parallel group, multicenter, seamless Phase 2b/3 program designed to evaluate the efficacy and safety of guselkumab in adult patients with moderately to severely active ulcerative colitis who experienced an inadequate response or who demonstrate intolerance to conventional therapy (e.g., thiopurines or corticosteroids), other biologics and/or JAK inhibitors (i.e., tumor necrosis factor [TNF]-alpha antagonists, vedolizumab, or tofacitinib). QUASAR includes a Phase 2b dose-ranging induction study, a confirmatory Phase 3 induction study, a Phase 3 randomized withdrawal maintenance study, and a long-term extension study through a total of 5 years.

In the induction trial, patients were enrolled if they had had an inadequate response to or could not tolerate conventional and/or advanced therapies. Patients were required to have a modified Mayo score (mMS) between 5 to 9, a rectal bleeding score of 1 or lower, and an endoscopy subscore no higher than 2. Patients (n=701) were randomized 3:2 to receive intravenous guselkumab (200 mg) or placebo at weeks 0, 4, and 8. The primary endpoint was clinical remission at week 12, defined as a Mayo stool frequency subscore of 0 or 1 with no increase from baseline, a rectal bleeding subscore of 0, and a Mayo endoscopy subscore of 0 or 1 with no friability. The trial met its primary endpoint, demonstrating clinical remission rates of 22.6% with guselkumab and 7.9% with placebo at week 12 ($p < .001$). In a comparison of the guselkumab arm and the placebo arm at week 12, the rates of symptomatic remission were 49.9% vs 20.7% ($p < .001$) and the rates of clinical response were 61.5% vs 27.9% ($p < .001$), respectively. Also at week 12, the rate of endoscopic improvement with guselkumab was superior to the rate with placebo (26.8% vs 11.1%; $p < .001$), as was the rate of histologic-endoscopic improvement (23.5% vs 7.5%; $p < .001$). Safety data were consistent with the known safety profile of guselkumab observed in patients with the approved indications.

The maintenance portion of the trial evaluated 568 patients who received one of two intravenous guselkumab induction regimens, including the recommended 200 mg regimen, for 12 weeks in the induction studies and demonstrated clinical response per mMS after 12 weeks. Patients were rerandomized to receive a subcutaneous (SC) maintenance regimen of either guselkumab 100 mg every 8 weeks, guselkumab 200 mg every 4 weeks, or placebo for up to an additional 44 weeks. In UC2, 42% of subjects had failed (inadequate response, loss of response, or intolerance) treatment with one or more biologics or JAK inhibitors. The primary endpoint was clinical remission at Week 44 defined by mMS. Secondary endpoints included corticosteroid-free clinical remission, endoscopic improvement, histologic endoscopic mucosal improvement, all at Week 44 and maintenance of clinical remission at Week 44 in subjects who achieved clinical remission 12 weeks after intravenous guselkumab induction treatment. The study found that 50% (n=190) of patients receiving guselkumab 200 mg SC maintenance every 4 weeks and 45% (n=188) of patients receiving guselkumab 100 mg SC every 8 weeks achieved primary endpoint of clinical remission at week 44 compared to 19% (n=190) placebo-treated patients ($p < 0.001$). Moreover, 34% (200 mg) and 35% (100 mg) of patients achieved endoscopic remission at one year with guselkumab maintenance therapy compared to 15% placebo-treated patients ($p < 0.001$).

Appendix

Examples of Clinical Reasons to Avoid Pharmacologic Treatment with Methotrexate, Cyclosporine, Acitretin, or Leflunomide

1. Clinical diagnosis of alcohol use disorder, alcoholic liver disease or other chronic liver disease
2. Drug interaction
3. Risk of treatment-related toxicity
4. Pregnancy or currently planning pregnancy
5. Breastfeeding
6. Significant comorbidity prohibits use of systemic agents (e.g., liver or kidney disease, blood dyscrasias, uncontrolled hypertension)
7. Hypersensitivity
8. History of intolerance or adverse event.

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The above policy is based on the following references:

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